

- 1 Resistance to temperature and temperature changes to ensure that the equipment doesn't fail.
- 2 Enough power to transmit signals to Earth or the Deep Space Network of satellites.



Voyager 1 is one of the most important space probes ever built

## Challenges in Upgrades

There are hardly any upgrade challenges at a hardware level because many a times, devices are replaced instead of upgraded. Yet, a system which is controlled by an embedded device can receive an external hardware upgrade, e.g. the camera of a phone can be replaced by service centre personnel. In situations where upgrades or hardware changes are required, the design must be highly modular and allow for the replacement of parts that are not embedded on the main board.

And then there are software upgrades. Think of your iPhone being upgraded from iOS5 to iOS6. Though this chapter covers hardware challenges, software goes hand-in-hand with hardware as far as enabling updates is concerned. For example, if the phone is to be upgraded to the next version but not beyond, the hardware included need not be very powerful, but just powerful enough to work for the anticipated improvements in the newer version of the software. The case with networking equipment is largely the same.

## Heat Dissipation

No matter what methods you use, as long as electronic equipment runs, it generates heat. While you can use a monster cooling fan for a gaming PC at home, embedded systems have limitations. Embedded systems can range from low power devices running on batteries to those powered by constant power supply from the wall socket.

Heat dissipation is a major issue for any system; if not handled properly, excess computation can melt down the system just by the heat it generates by itself. In certain cases, where there is enough space, fans can be used. However, equipment designed to be small, silent or powered by a battery can't use the technique and must find other ways to stay cool. Punching holes at the top of the body to let hot air escape is one of the techniques commonly used. Systems that need to be as compact as possible (such as a mobile phone) must use better heat conducting materials and design, and ideally radiate as much heat as possible through the outer casing.